

Orion Nichols

Mechanical Engineer | www.linkedin.com/in/orionnichols | onichols0722@gmail.com | (530)-601-1688

Education	Cal Poly Humboldt / Mechanical Engineering B.S. Spring 2026 - GPA 3.4 Capstone Autonomous Robotic Pruning System Using Bio-Inspired Locomotion
Work & Research Experience	STAD Research Group / Student Research Assistant Fall 2024 - Present - Research on thermal necrosis due to drilling operations into bone Cal Poly Humboldt Student Learning Center / Tutor Fall 2024 - Tutored students on CAD fundamentals in both SolidWorks and AutoCAD - Assisted students with their Solid Mechanics and Statics/Materials homework JLSky Consultant / Engineering Design Consultant June 2024 - August 2024 - Designed HVAC and P&ID systems using AutoCAD for commercial use - Assisted in the design of the Kabobinator, an automated kabob-making machine J.C. Chang & Associates / Mechanical Engineering Intern April - August 2023 - Designed HVAC layouts for commercial buildings in AutoCAD - Completed job walks for ongoing projects and project bids - Collected information to make informed design decisions and assist with project proposals SunPower R&D / Mechanical Engineering Intern May 2020 - Oct. 2020 - Researched effects of different weather environments on solar module efficiency - Conducted study on viability of utilizing microinverters for power generation Starbucks / Shift Manager 2019 - Feb. 2023 - Assistant store manager in charge of scheduling, inventory, and planning Citrus Circuits FRC Robotics Team / Mechanical Fabrication August 2015 - June 2018 - Operated and trained new members on various heavy machinery, including lathes, mills, and CNC machinery, as well as shop safety, lab usage, hand tools, and CAD software
Extra Curriculars	Aviation Club / President Spring 2024 to Spring 2025 - Assisted with installation and setup for the school's new Advanced Aviation Training Device ASME Chapter / President Fall 2025 - Managed the transition from startup local chapter to established community on campus COMAP Math Modeling Competition / Participant Spring 2026 - Developed a mathematical model to create practical phone battery load shapes
Skills	CAD: SolidWorks, Fusion 360, AutoCAD, Bluebeam Revu, G-code Software: MATLAB, Python, Java, Microsoft/Google Suite, Mathematica Machinery: lathe, mill, CNC routers, 3D printing, Keyence 3D microscope, dynamometers